Vipul Sapra

Charlotte, North Carolina | 513-693-6897 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Databricks Lakehouse Architecture, Snowflake Dimensional Modeling, Cloud Data Warehousing, Distributed Computing, Agentic AI Workflows, LLM Prompt Orchestration, Autonomous Anomaly Detection, MLOps, Predictive Modeling, Time-Series Forecasting, ETL/ELT Pipeline Optimization, Data Governance & Lineage.

Programming Languages & Tools: Databricks, Snowflake, AWS (S3, Redshift), Azure (ADF, Databricks, Synapse, SQL), Apache Spark, PySpark, Python, XGBoost, Scikit-Learn, SARIMA, Power BI, Tableau, Collibra, KIRO IDE, GitHub Copilot.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Lead Cloud Data & AI Software Engineer — **May 2022 – Present**

- **Architected and deployed distributed cloud data platforms** utilizing Databricks, Snowflake, AWS (S3, Redshift), and PySpark to process hundreds of terabytes with 99.9% data availability SLAs, directly aligning with JPMorganChase's high-throughput infrastructure scale.
- **Engineered custom autonomous agentic AI workflows** and prompt orchestration frameworks using KIRO IDE, Copilot, and LLMs to automate schema discovery, data profiling, and code migration, reducing manual delivery turnaround by 40-50%.
- **Deployed autonomous machine learning anomaly detection systems** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, safeguarding downstream BI pipelines and institutional data integrity.
- **Built and scaled predictive machine learning models** using XGBoost, Scikit-Learn, and Advanced Time-Series (SARIMA) for customer behavior forecasting and workforce capacity optimization, improving service reliability and reducing response times by 20%.
- **Streamlined conversational AI testing and enterprise data lineage** by engineering automated evaluation frameworks for SpectrumIQ and implementing Collibra data governance firewalls, reducing operational turnaround by over 80%.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Designed production-grade machine learning classification pipelines** leveraging XGBoost and Random Forests to predict enterprise customer behavior and deliver actionable predictive insights.
- **Implemented robust time-series forecasting models** using SARIMA algorithms to predict field support demand and truck rolls, optimizing enterprise capacity allocation.
- **Optimized cloud ETL performance and pipeline reliability** across Azure Databricks, Synapse, and SQL ecosystems to support advanced predictive analytics workflows.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Developed executive Key Operating Metric (KOM) dashboards** and semantic models in Power BI using DAX to translate complex telemetry into corporate strategy.
- **Engineered source-to-target reconciliation firewalls** to guarantee absolute data accuracy and compliance across critical enterprise reporting layers.
- **Created interactive Tableau visualizations** empowering executive leadership with real-time operational insights and data-driven decision making.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.